



Note: To install using the source files, download the latest source tar ball from CouchDB's official site: <http://couchdb.apache.org/#download>. Once downloaded, uncompress it and build the packages using the *make* command.

Running basic commands

After a successful installation, let's check whether our CouchDB instance is up and running. For that, simply run a *curl GET* command as shown:

```
# curl -X GET http://<COUCHDB_SERVER_IP>:5984
```

...where *COUCHDB_SERVER_IP* will be the IP address of the computer on which you have installed CouchDB.

You should get the following output as a result of the above command:

```
{"couchdb":"Welcome","uuid":"e833c5cfb43d9601433cae620e74595","version":"1.6.1","vendor":{"name":"The Apache Software Foundation","version":"1.6.1"}}
```

Next, let's create a simple database using the *curl PUT* parameter:

```
# curl -X PUT http://<COUCHDB_SERVER_IP>:5984/<DATABASE_NAME>
```

You should get the following response:

```
{"ok":true}.
```

This means you have successfully created a new database on CouchDB.

While naming your database, always remember the following useful points:

- Use lowercase characters (a-z)
- You can use digits (0-9) while naming your database but the first character has to be a letter of the alphabet
- You can use any of the characters `_`, `$`, `(`, `)`, `+`, `-` and `/` while naming your database but the first character has to be a letter of the alphabet

You can obtain additional details about your database by simply querying against it. Once again, let's use the *curl GET* command to pull the relevant database information:

```
# curl -X GET http://<COUCHDB_SERVER_IP>:5984/<DATABASE_NAME>
```

This provides a detailed description of your database as shown below:

```
{
  "db_name":"<DATABASE_NAME>",<br>  "doc_count":0,
```

```
[root@DB-HOST-1 ~]#
[root@DB-HOST-1 ~]# curl -X GET http://192.168.0.40:5984/_all_dbs
[{"replicator": "users", "oranges": "stuff"}]
[root@DB-HOST-1 ~]#
[root@DB-HOST-1 ~]#
```

Figure 1: Output of all databases

```
  "doc_del_count":0,
  "update_seq":0,
  "purge_seq":0,
  "compact_running":false,
  "disk_size":79,
  "data_size":0,
  "instance_start_time":"142779596351454",
  "disk_format_version":6,
  "committed_update_seq":0
}
```

CouchDB also allows you to list all your databases in one go using some special built-in commands, such as *_all_dbs*.

```
# curl -X GET http://<COUCHDB_SERVER_IP>:5984/_all_dbs
```

The output in Figure 1 shows you a list of all the databases present in this particular CouchDB instance. In the output, you will also notice the presence of two CouchDB internal databases/interfaces called *_users* and *_replicator*. These interfaces are managed and run by CouchDB itself. A brief list of these interfaces is given below:

- *_users*: Contains all CouchDB users
- *_active_tasks*: Lists all actively running tasks in the database
- *_db_updates*: Lists all current occurring database events
- *_log*: Displays CouchDB's internal logs
- *_replicate*: This database is primarily used during the database replication process
- *_restart*: Restarts the CouchDB instance
- *_stats*: Displays the database as well as CouchDB server's statistics
- *_utils*: Used to provide access to the built-in CouchDB administration interface
- *_uuids*: Used to request one or more uniquely generated UUIDs from CouchDB
- *_config*: Displays the entire CouchDB configuration details

Now that our database is ready, let us populate it with a document. As we discussed earlier, documents are the primary unit of data in CouchDB and consist of any number of fields and attachments. Documents also include metadata that's maintained by the database system. Document fields are uniquely named and contain values of varying types (text, number, boolean, lists, etc), and there is no set limit to the text size or element count.

To add a document to a database, let's use the *curl*